

Introduction of a SaMD : VeriOsteo[®] and Its clinical Impact

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Disclosure

- Director of Orthopedic Department
- Executive Director of Smart Healthcare Committee
- Executive Director of Clinical Informatics Committee

Education

- Med. student of National Yang-Ming Medical College
- PhD of Biomedical Engineering, National Yang-Ming Univ.

Skills

- Spinal Surgery including ALIF/OLIF/xLIF, MISS, tumors operation (TES) and deformity correction (PSO, SPO, pVCR)
- Arthroplasty Surgery
- Orthopedic Robotic Surgery
- Medical Informatics and biomedical engineering



Introduction

- Osteoporosis is the 2nd leading major disease of the elderly around the world
- The 4th most common chronic disease for the aged people older than 65 y/o in Taiwan
- an osteoporotic fracture occurs every 3 seconds worldwide



Worldwide,
an osteoporotic
fracture occurs
every 3 seconds¹

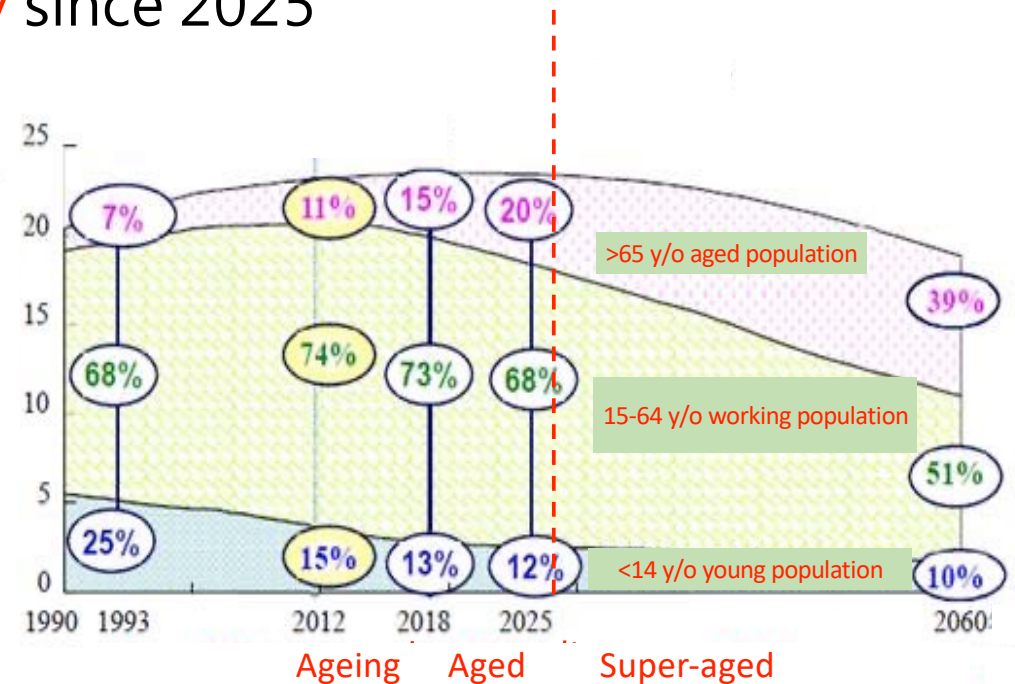
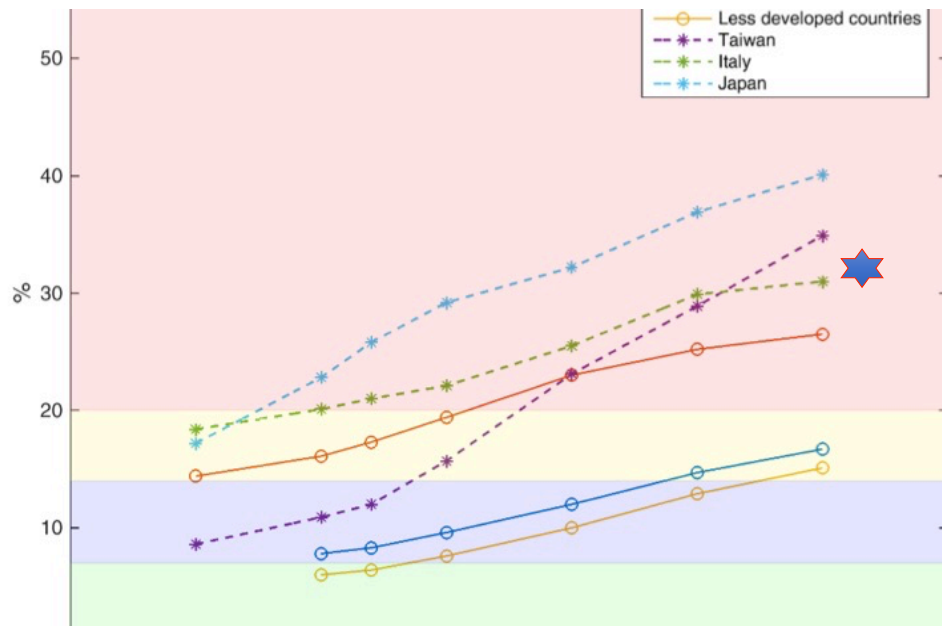
1. International Osteoporosis Foundation (IOF). Facts and statistics. Available at <https://www.iofbonehealth.org/facts-statistics> Accessed September 23, 2017.

Introduction

- Prevalence of osteoporosis in women : 30%
- Lifetime incidence of osteoporotic fracture
 - 1/3 for woman
 - 1/5 for man
- 1/2 will get another osteoporotic fracture

Introduction

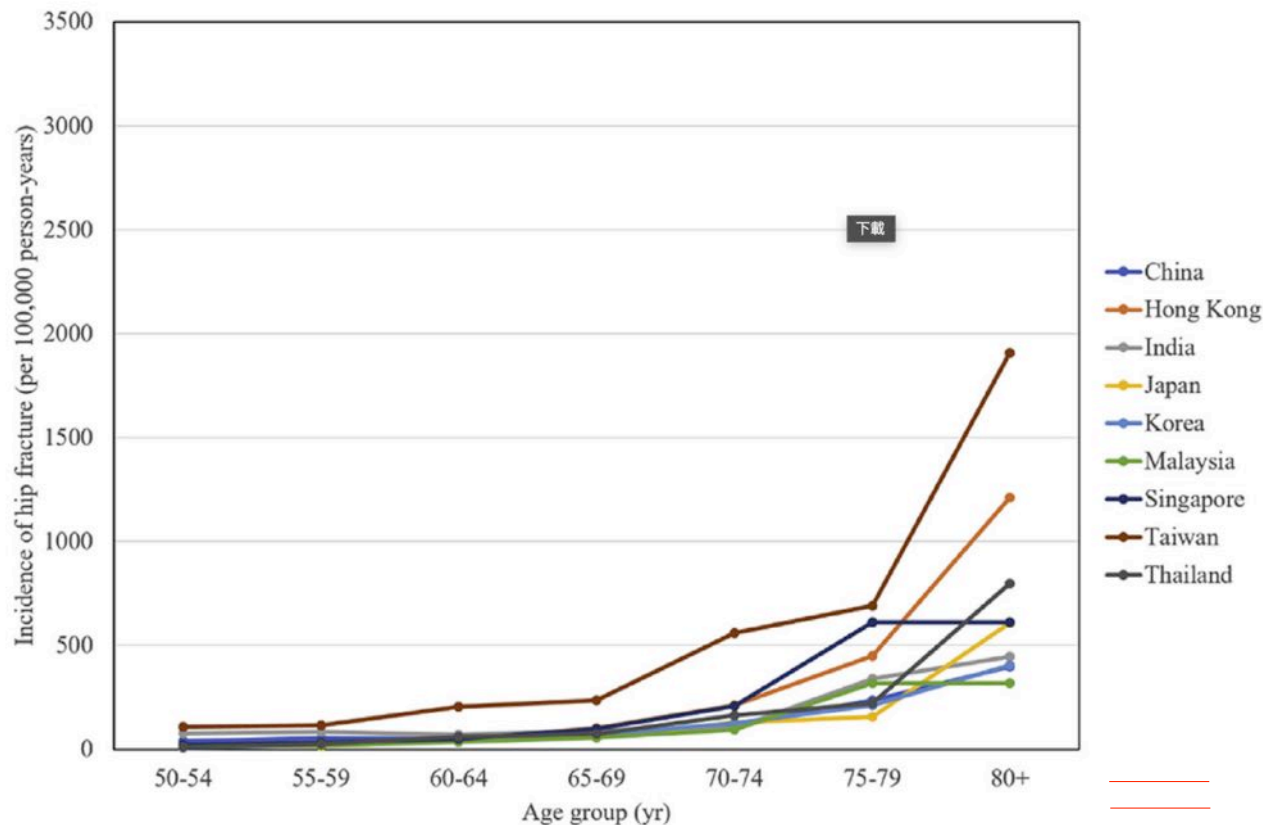
- Taiwan is a **super-aged society** since 2025
- Japan since 2007



1990 - 2060 Population shifting and projection of Taiwan

Introduction

- Hip fracture : highest in Asia, 9th around the world



C.-L. Cheung et al. / Osteoporosis and Sarcopenia 4 (2018) 16e21

Table 1: Age-standardized hip fracture rates (per 100 000 population) across different continents

Continent	Country	Men	Women
Asia	Morocco, Rabat	57.7	79.9
	Cameroon	43.7	52.1
	China, Beijing	87	97
	China, Shenyang	101.3	80.9
	Korea	137	262
	Iran	127.3	164.6
	Malaysia	87.4	212.5
	Japan, Tottori	107.3	297.3
	Japan	99.6	368
	Kuwait	216.6	316
	Singapore	152	402
	Hong Kong	193	484.3
	Hong Kong	50	110
	Taiwan	233.4	496.8
South America	Mexico	98	169
	Brazil, Sobral	59.3	168.4
	Argentina	137	405
	Venezuela	37	98
Europe	Switzerland	137.8	346
	Former East Germany	137.8	354.7
	Former West Germany	154.5	399.4
	England	143.6	418.2
	Greece	201.7	469.9
	Sweden	302.7	709.5
	Norway	352	763.6
	Norway, Oslo	399.3	920.7
	Austria	567	759
	Hungary	223	430
North America	The Netherlands	308	669
	United States, Minnesota	201.6	511.5
	United States	197.2	553.5
Oceania	New Zealand, Maori	197	516
	Non-Maori	288	827
	New South Wales	191.8	475.1
	Australia	187.8	504.2
	Australia	130	390

Diagnosis of osteoporosis

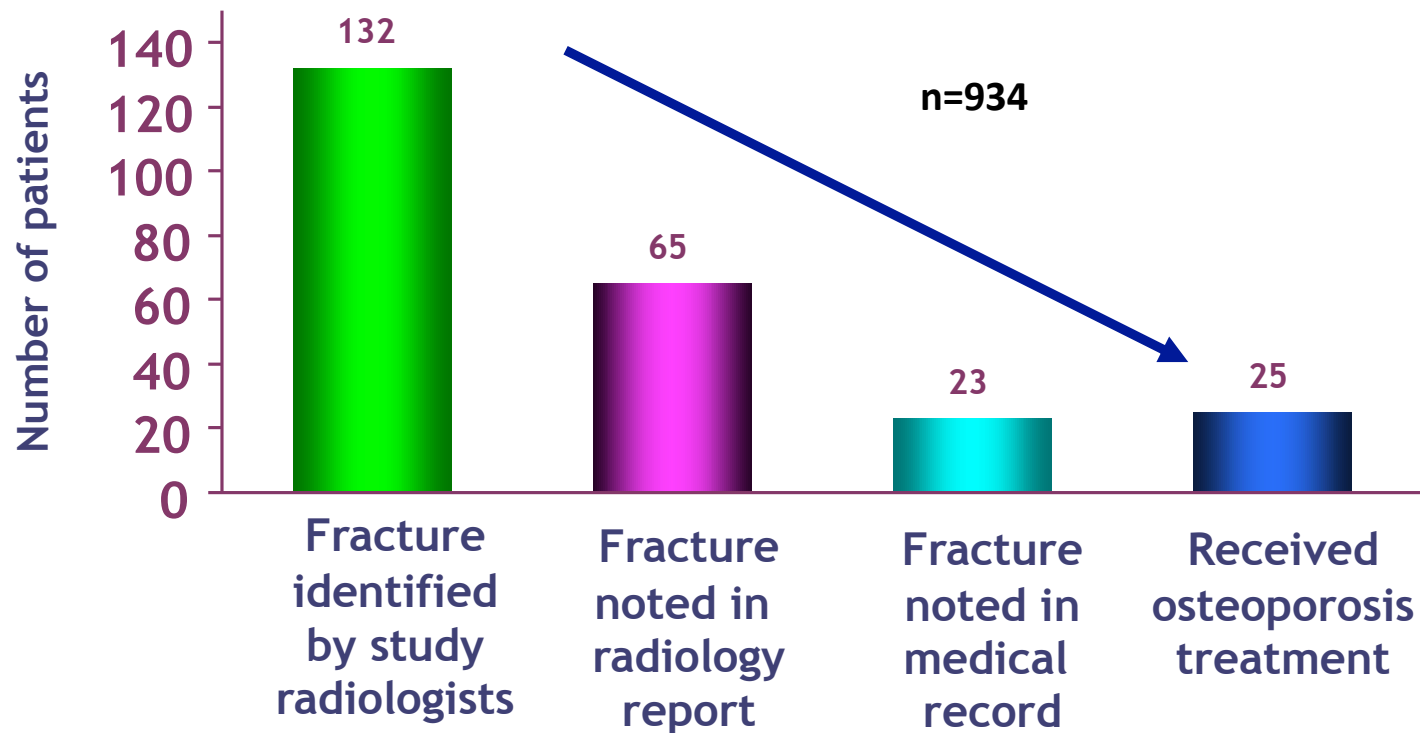
- **Clinical osteoporosis** : low energy trauma induced fracture
- By **DXA** (Dual-energy X-ray absorptiometry)
 - T-score : Difference expressed as standard deviation compared to young (20's) reference population
 - Normal : ≥ -1.0
 - Osteopenia : $-1.0 \sim -2.5$
 - **Osteoporosis** : ≤ -2.5
 - Severe Osteoporosis : Osteoporosis with one or more osteoporotic fractures

Optimal Treatment for osteoporotic fracture

Prevention

Osteoporosis Often Underwent Undetected

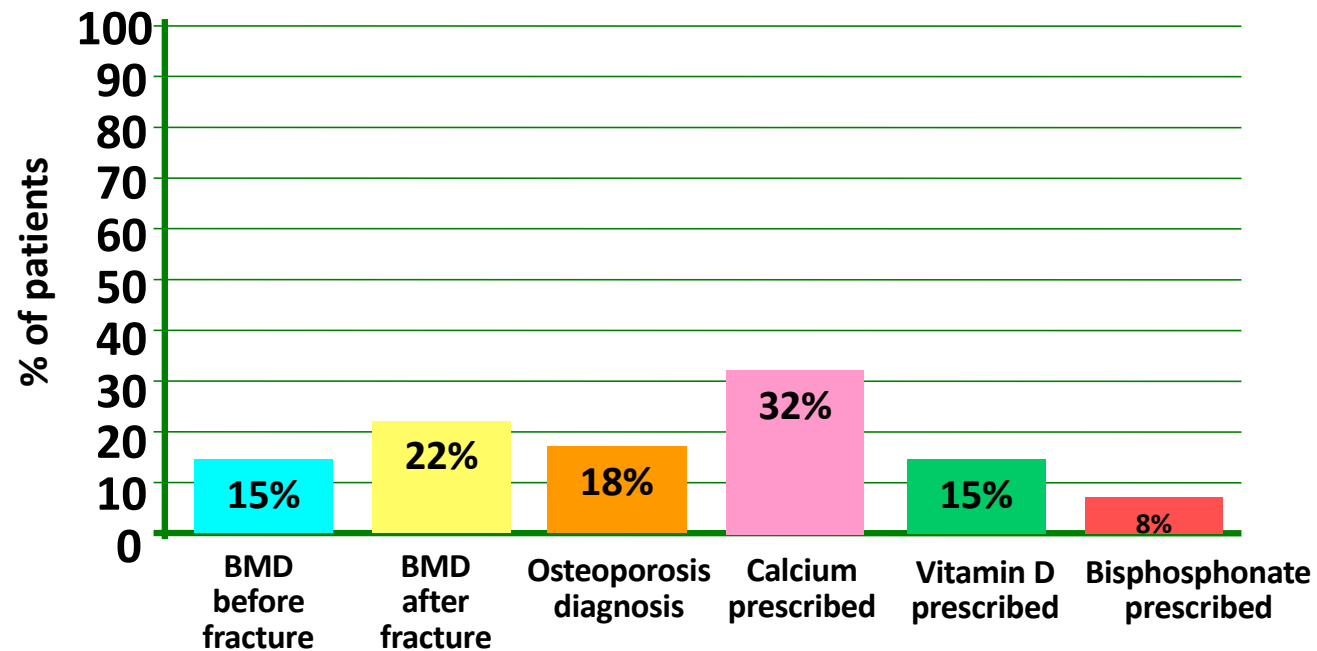
Only 19% Of Women With A Fracture Received Treatment



Gehlbach SH et al. Osteoporos Int. 2000;11:577-582.

Undiagnosed and Untreated

Patience Admitted with Osteoporotic Fractures



Data from Hajcsar EE, et al. *CMAJ*. 2000;163:819-822.

Prevalence

Male : 23.9%
Female : 38.3%



Huge Gap

Detection Rate

Male : 1.6%
Female : 11.3%

Taiwan Data from 2004 to 2008

Clinical Unmet Need

Easy applied detection tools



Solution

CXR → Bone Mineral Density(DXA)

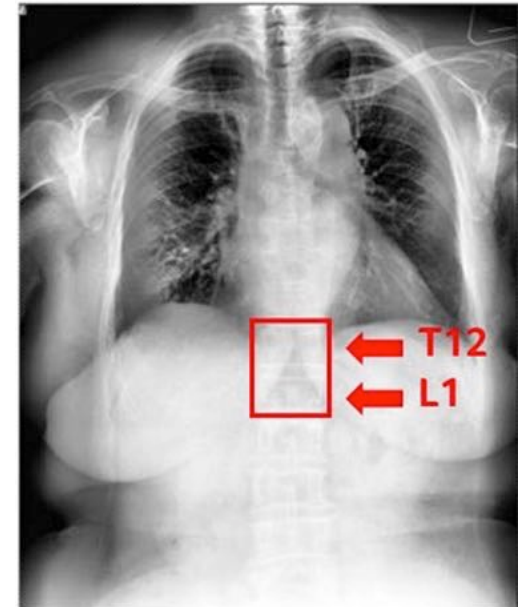
Reference: Yen, CC., Lin, WC., Wang, TH. *et al.* Pre-screening for osteoporosis with calcaneus quantitative ultrasound and dual-energy X-ray absorptiometry bone density. *Sci Rep* 11, 15709 (2021). <https://doi.org/10.1038/s41598-021-95261-7>

Objective

- An easily applied detection tool
- The Chest X-ray (CXR) , PA view
 - Standard of routine physical examinations
 - The most frequently used diagnostic image modality
- By deep learning model, to estimate bone mineral density (eBMD) from CXR PA view image, and to predict the possibility of osteoporosis

Materials and Methods

- ROI (region of interest) = T12 and L1
 - Osteoporotic fractures commonly occurred
- To identify Individuals with high risk of osteoporosis (eBMD T-score ≤ -2.5)
- In collaboration with Acer Medical Inc.
- AI Model name: VeriOsteo® OP



Materials and Methods

Training Data Set

- Included Criteria
 - From VGHTC, 2021-2022
 - Age ≥ 50
 - With both CXR and DXA
 - Interval ≤ 6 months

- Excluded Criteria

- Lack any of L1, L2, L3 or L4 value on BMD
- T-score gap between adjacent vertebrae ≥ 1.0

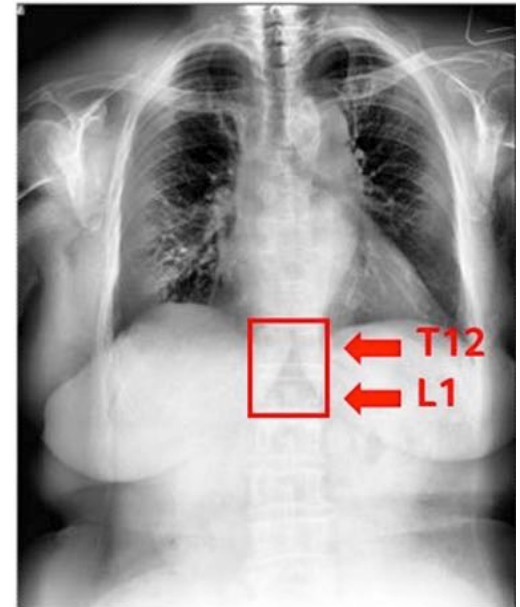
Testing Data Set

- From VGHTC, 2021-2022
- From Joy Clinic, 2012-2021

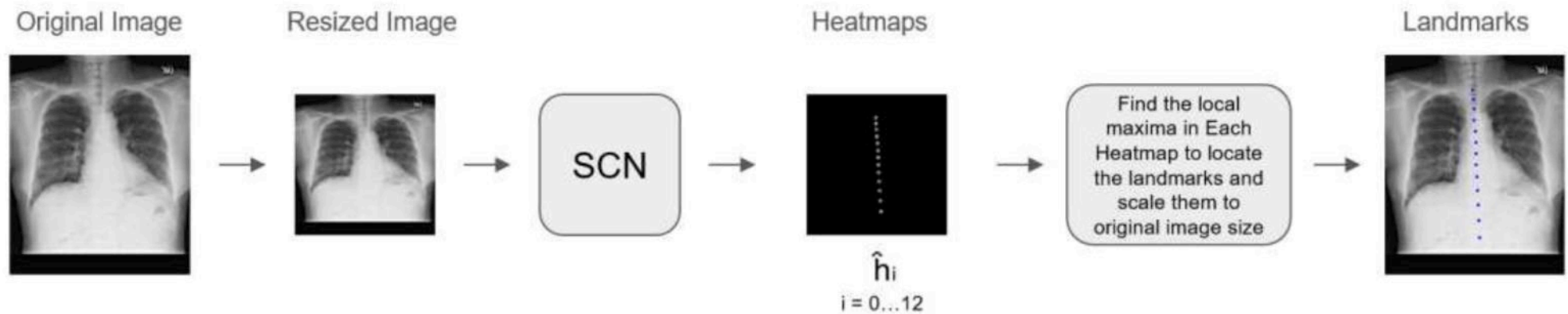
	CXR Quantity
Training/Validating	5122 (VGHTC)
Testing	440 (VGHTC + Joy Clinic)

Image Pre-processing

- CLAHE : contrast limited adaptive histogram equalization algorithm, a sophisticated method used widely in medical imaging to **improve contrast** while **retaining essential details** within the images

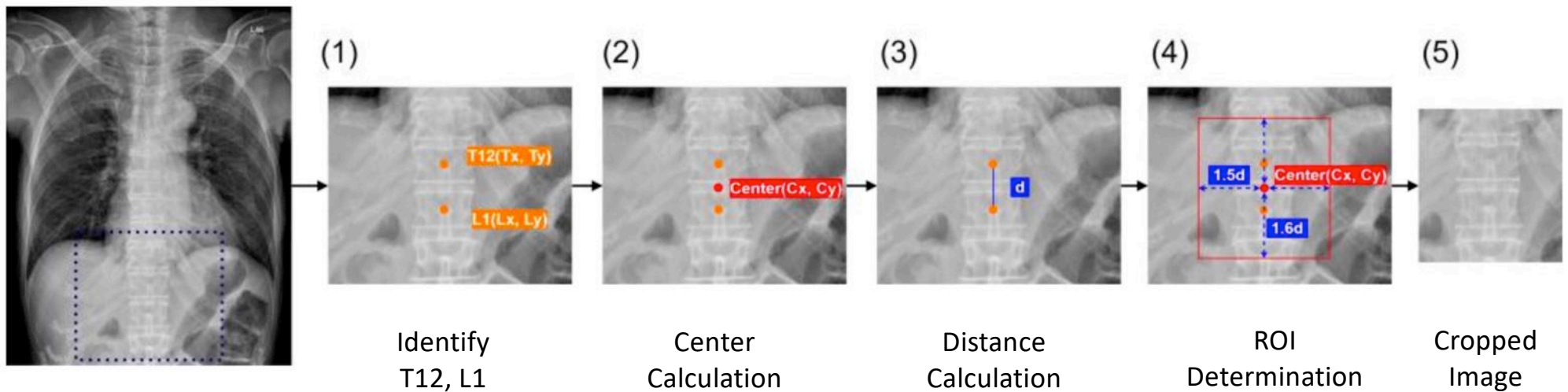


SCN (Spatial Configuration-Net) module



The spinal detection module identifies vertebrae within thoracic cavity X-ray images using the SCN (Spatial Configuration-Net) module, which **outputs a heatmap indicating the position of each vertebra**

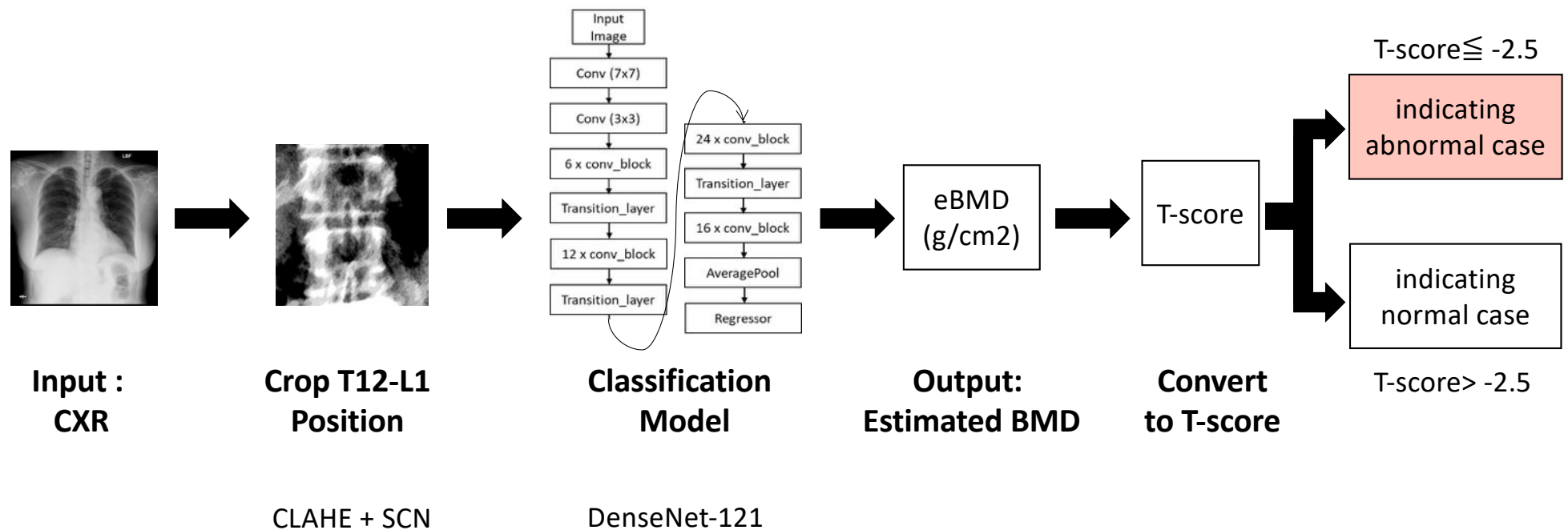
T12-L1 Crop module



BMD Prediction DL Model

- **Pilot study** with 2002 training samples :
 - VGG-16, DenseNet-121, ResNet-50, and EfficientNetV2S
 - Area Under the Curve (AUC) as the primary metric for model selection
- The AUC results
 - VGG-16 (0.89)
 - ResNet-50 (0.90),
 - EfficientNetV2S (0.90)
 - **DenseNet-121 (0.92)**

Methods – Prediction Process



- CLAHE : Contrast Limited Adaptive Histogram Equalization
- SCN : SpatialConfiguration-Net

Results

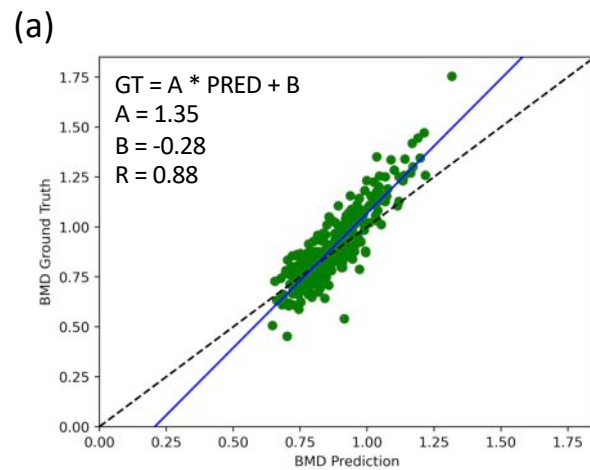
	VGHTC	Joy Clinic	Overall
Accuracy (95% CI)	88.33% (84.15% to 91.74%)	90.44% (84.21% to 94.81%)	88.99% (85.67% to 91.77%)
Sensitivity (95% CI)	86.74% (80.92% to 91.32%)	94.03% (85.41% to 98.35%)	88.71% (84.10% to 92.37%)
Specificity (95% CI)	90.76% (84.06% to 95.29%)	86.96% (76.68% to 93.86%)	89.36% (84.05% to 93.38%)
PPV** (95% CI)	93.45% (88.59% to 96.69%)	87.5 % (77.59% to 94.12%)	91.67% (87.42% to 94.84%)
NPV** (95% CI)	81.82% (74.17% to 87.99%)	93.75 % (84.76% to 98.27%)	85.71% (80.02% to 90.29%)

** PPV : Positive Predicted Value, NPV : Negative Predicted Value

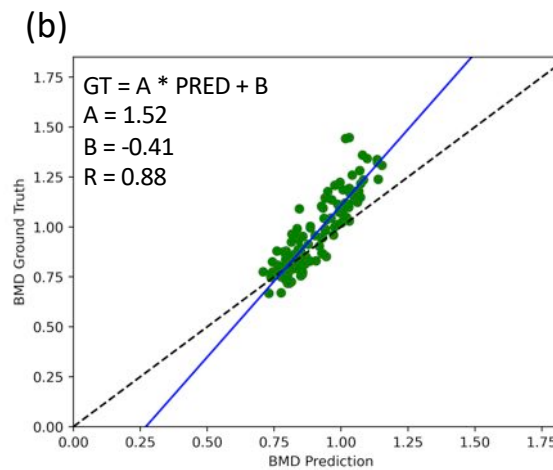
Higher sensitivity is better due to clinical unmet need
Cross institutional validation didn't decrease its sensitivity

Results – BMD

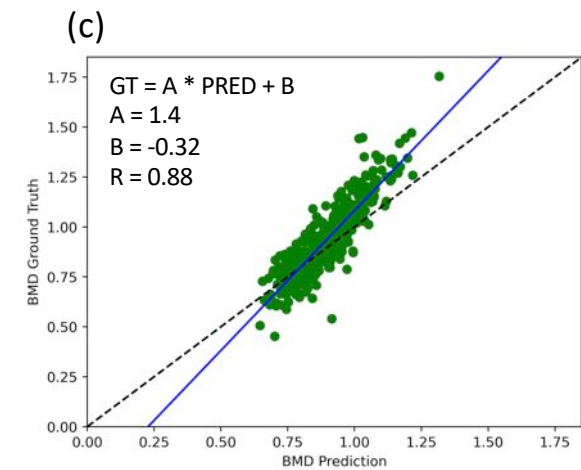
- Model-predicted-BMD and BMD-from-DXA “gold standard” :
 - Significant correlation ($R^2 = 0.88$)



VGHTC



Joy Clinic

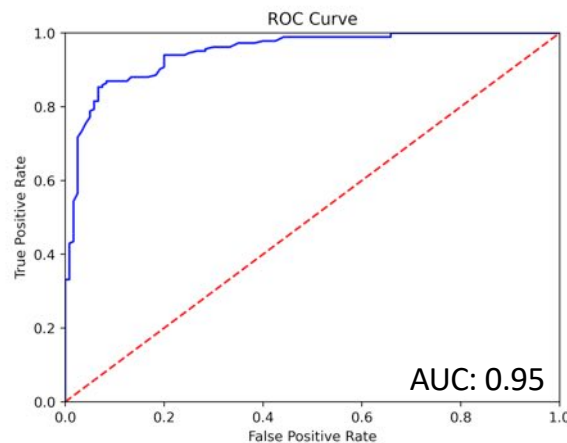


Overall

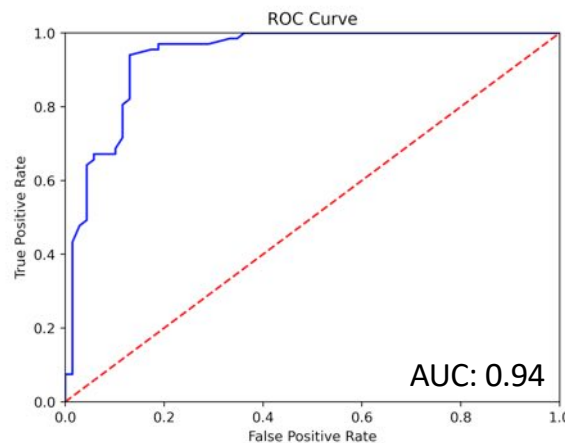
Results – Osteoporosis prediction

- Osteoporosis prediction
 - Positive = T-score ≤ -2.5
 - Negative = T-score > -2.5

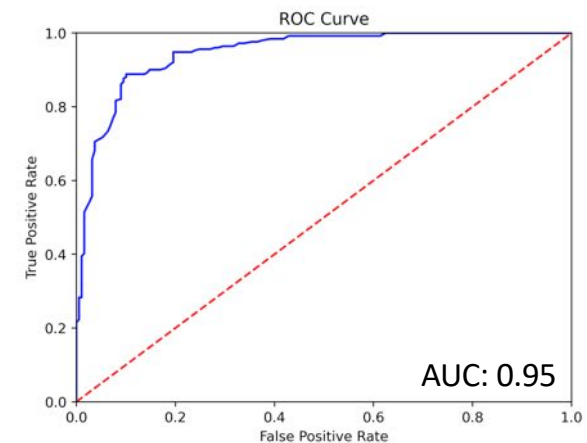
	VGHTC	Joy Clinic	Overall
AUC (95% CI)	94.82% (92.4%, 97.2%)	93.75% (89.5%, 98%)	94.61% (92.5%, 96.7%)



VGHTC



Joy Clinic



Overall

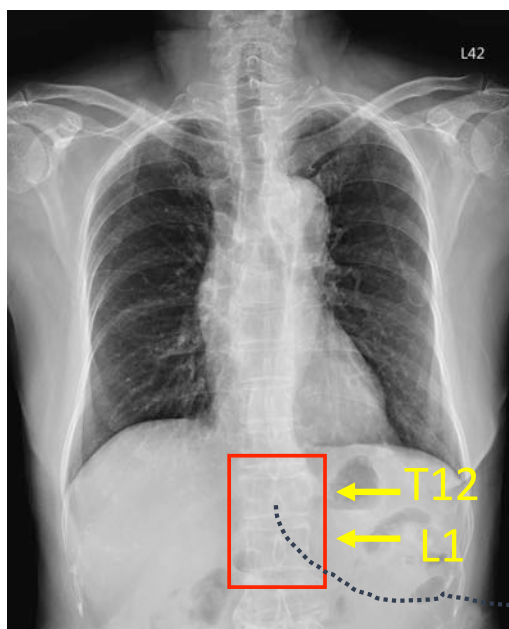
Fig. The Receiver Operating Characteristic (ROC) curve analysis was conducted to assess the performance of the model. (a) VGHTC, (b) Joy Clinic, (c) Overall performance (VGHTC and Joy Clinic combined).



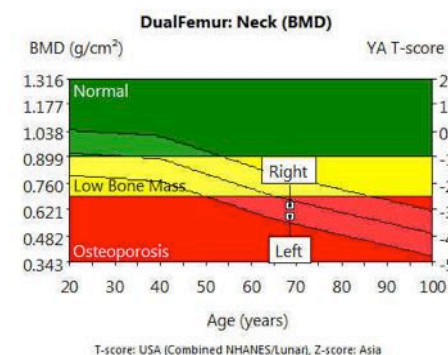
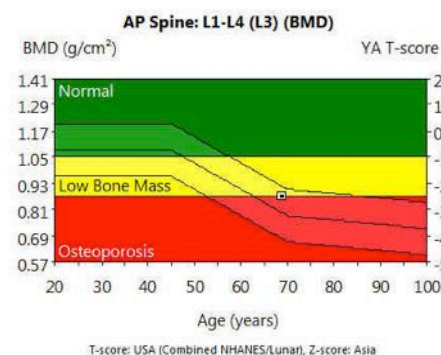
T-FDA certified **VeriOsteo OP** SaMD in 2023/11



CXR-PA (T12, L1) L1-L4 BMD Osteoporosis or not



AI Predict
L1-L4 BMD



Region	BMD (g/cm ²)	YA T-score	AM (%)	AM Z-score	WHO Classification
AP Spine					
L1-L4 (L3)	0.872	-2.5	109	0.6	Osteoporosis
DualFemur					
Neck Left	0.587	-3.2	87	-0.7	Osteoporosis
Neck Right	0.646	-2.8	96	-0.2	Osteoporosis
Total Left	0.698	-2.5	93	-0.4	Osteoporosis
Total Right	0.708	-2.4	95	-0.3	Low Bone Mass

Discussions – Osteoporosis prediction

- FRAX for osteoporosis predictions on 50-65 y/o female
 - AUC : 0.58
 - US Preventative Services Task Force FRAX threshold has a low sensitivity to detect osteoporosis in women ages 50–64 years, S. Bansal, et al, Osteoporosis Int (2015) 26:1429–1433
- Our AI prediction model can provide
 - High AUC : 94.61%
 - Sensitivity : 88.71%, specificity : 89.36%, PPV : 91.67%, NPV : 85.71%

Multi-Center AI Model Performance

The clinical performance of ©VeriOsteo OP:

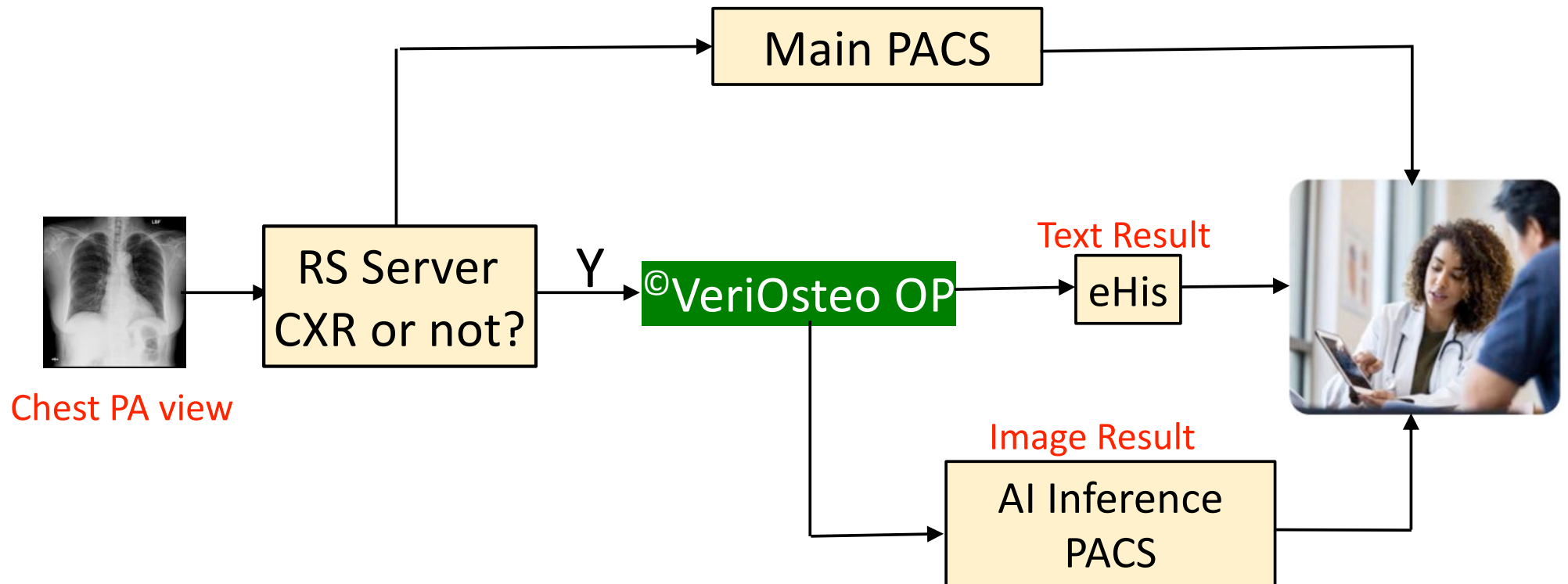
DXA as a Ground Truths	VeriOsteo OP [Clinical study: 2 center]	QUS** (Taiwanese population: single center)
Sensitivity	88.7%	67.2%
Specificity	89.4%	64.9%
Accuracy	89%	66.2%

[Acer Medical-Clinical study, 2023]

[Wei-Chun Lin et., al., 2021]

** QUS : Quantitative Ultrasound

Clinical process integration of ©VeriOsteo OP



Product summary



- ©VeriOsteo OP represents a promising and reliable auxiliary tool for the Osteoporosis prediction and warning
 - Using the T12, L1 image on CXR.
 - Validated by multicenter data.
- Provides an an opportunity for early detection of osteoporosis, with acceptably high AUC, PPV/NPV can provide osteoporosis early warning and related fracture prevention

Post-hoc application

- In 2023, at VGHTC, IRB proven post-hoc study, by ©VeriOsteo OP
 - A total number of 56,263 subjects, 91,507 Chest PA view performed
 - Including criteria : Female ≥ 50 y/o, male ≥ 65 y/o
 - Excluding criteria :
 - BMD performed within 2 years
 - deceased
- Result :
 - 16.2%(9,157 subjects) were found to be in risk of osteoporosis who is escaped from current OP screening system
 - 10% (9,157 from 91,507 images) capture rate per CXR
 - In patients with multiple evaluations : SD of T-score : about 0.18

GE vs Hologic

- Hologic and GE are both leaders in DXA imaging
- **Image resolution** : Hologic offers superior, high-resolution imaging
- **Radiation Dose** : GE Prodigy systems provide lower patient dose
- **Precision**: GE Prodigy may have lower precision errors in the spine and femur
- However, **their results are not directly interchangeable**, and requiring specific conversion equations

Anything wrong between GE vs Hologic ? **No...**

A Performance Evaluation in a 2023 Single-Center Cohort

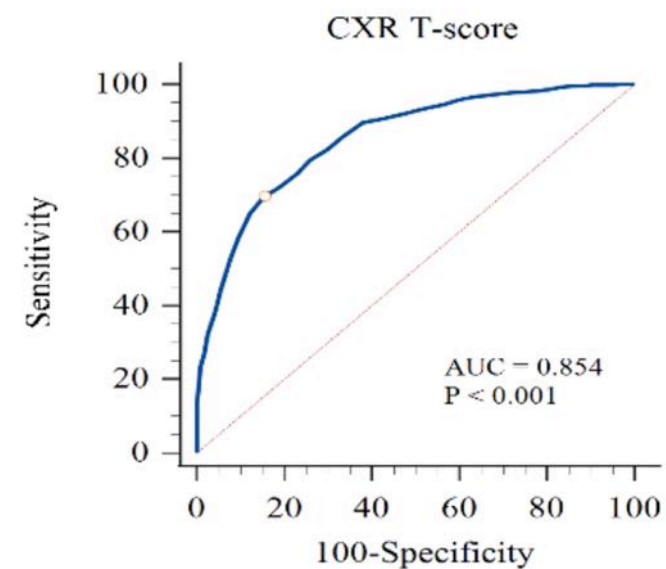
- Independent validation on Taichung Veterans General Hospital
- Unlike the previous validation, new cohort composed by 2,336 patients, with mean age of 69.9 ± 10.4 years
- **72.3%** (1688/2336) **are male**, serving as a **critical 'stress test'** to evaluate the model's robustness across biological sexes within a real-world clinical setting.
- Independent validation utilized **Hologic DXA** scanners in a veteran population.
 - Overall accuracy : 76.97%
 - AUC : 0.854

Clinical Performance and its Limitation Study

- 2,336 patients (72.3% male) from a medical center in 2023
 - Underwent both CXR and DXA.
- A independent validation utilized **Hologic DXA** scanners in a veteran population.
 - Overall accuracy : 76.97%
 - AUC : 0.854

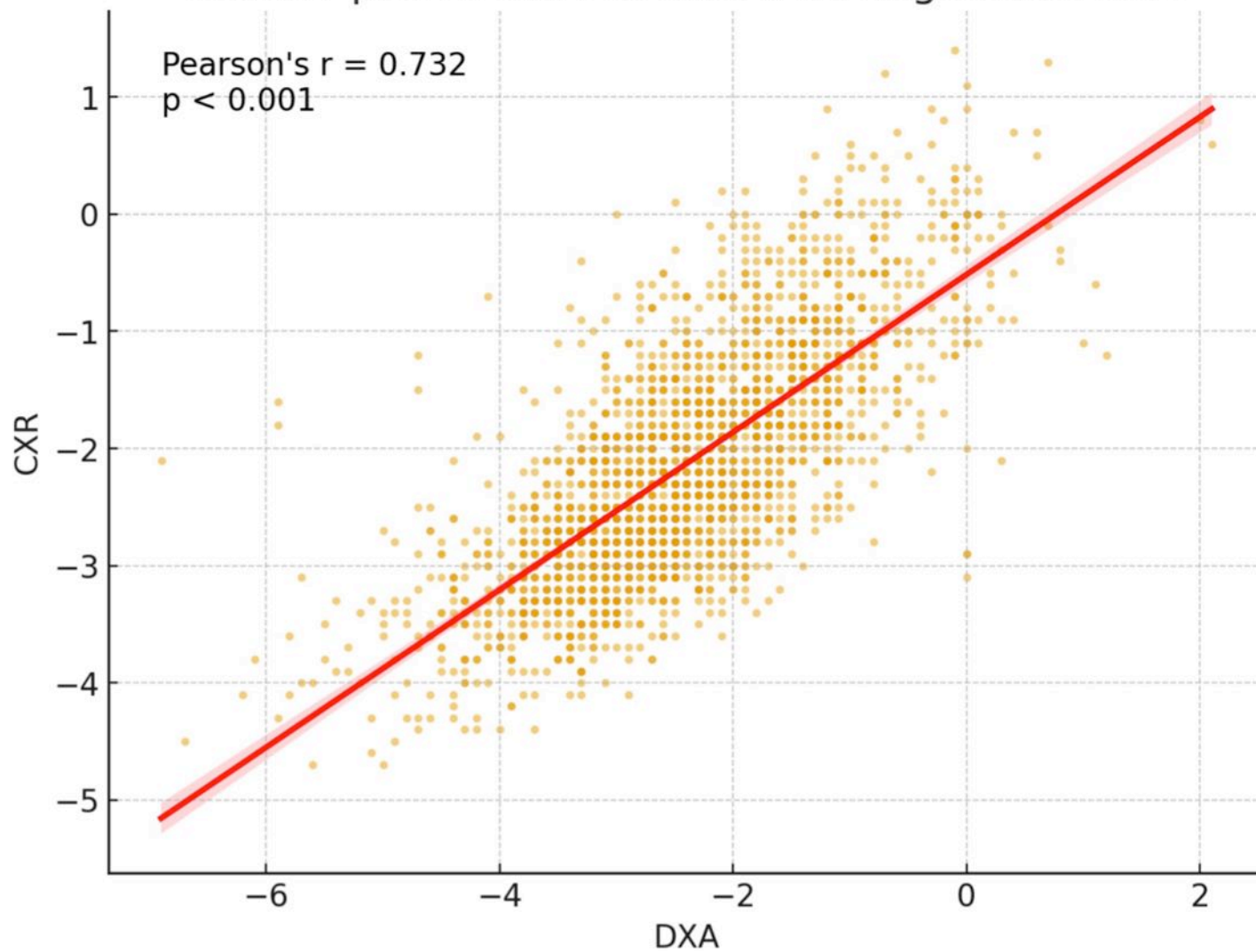
Table 2: Diagnostic Performance of CXR AI Model for Osteoporosis Detection (DXA T-score ≤ -2.5 as Reference)

Variables	Value (%)
Sensitivity	69.67%
Specificity	84.64%
Positive Predictive Value (PPV)	82.66%
Negative Predictive Value (NPV)	72.65%
Accuracy	76.97%
AUC(95% CI)	0.854(0.839-0.868)



AUC reduced from 0.945 to 0.854

Scatter plot of DXA vs CXR with Regression Line



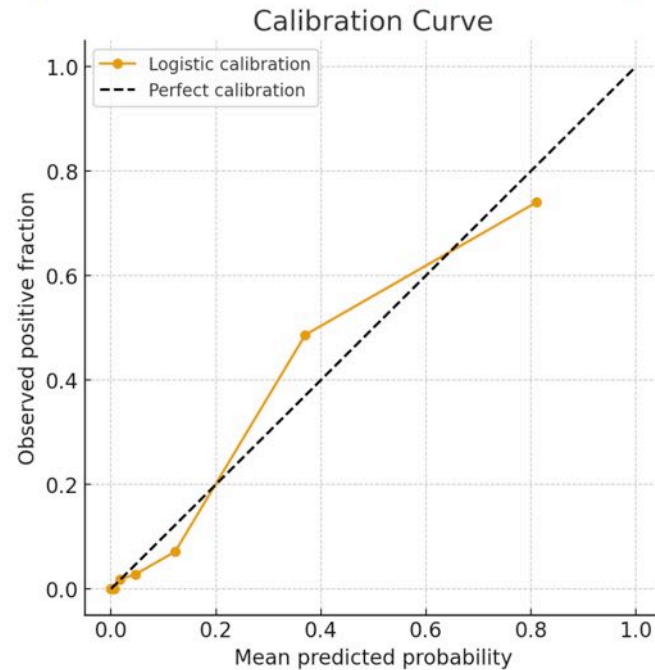
Discussion

- Sato et al. (2022) reported a correlation coefficient (R) of 0.63 for lumbar BMD prediction using chest X-rays combined with age and sex
- **Index study** of VeriOsteo® by our team and Acer Medical® showed Pearson's r of 0.88
- Our **stress test** study showed Pearson's r of 0.732 even in disparities in Cohort Characteristics and Domain Shift from female to male

Table 3. Subgroup Analysis of Diagnostic Accuracy for Osteoporosis by Gender

Gender	AUC (95% CI)	p-value	Optimal Cutoff	Sensitivity (%)	Specificity (%)	Accuracy (%)	PPV (%)	NPV (%)
Male	0.825 (0.794–0.854)	<0.001	≤ -1.9	73.58	76.15	75.31	60	85.57
Female	0.849 (0.831–0.866)	<0.001	≤ -2.5	74.92	79.09	76.66	83.39	69.24

Figure 5. Calibration curve of AI-predicted osteoporosis risk.



Advancing Diagnostic Equity Through Artificial Intelligence Chest radiograph screening for osteoporosis in asian populations

Chen, Shu-Han, et al. "Advancing diagnostic equity through artificial intelligence chest radiograph screening for osteoporosis in Asian populations." *npj Digital Medicine* (2026). (IF=15.1)

Figure 5. Calibration plot comparing predicted probabilities from the AI model with the observed prevalence of DXA-confirmed osteoporosis. The dashed diagonal indicates perfect calibration. Points represent decile-based bins; vertical bars denote 95% confidence intervals. The model showed good calibration (Brier score = 0.051; intercept = 0.068; slope = 1.089), with only minor deviations at the extremes.

The Calibration curve of VeriOsteo®

Figure 2A–C. Correlation between AI-predicted and DXA-measured bone mineral density (BMD) across BMI, age, and sex strata.

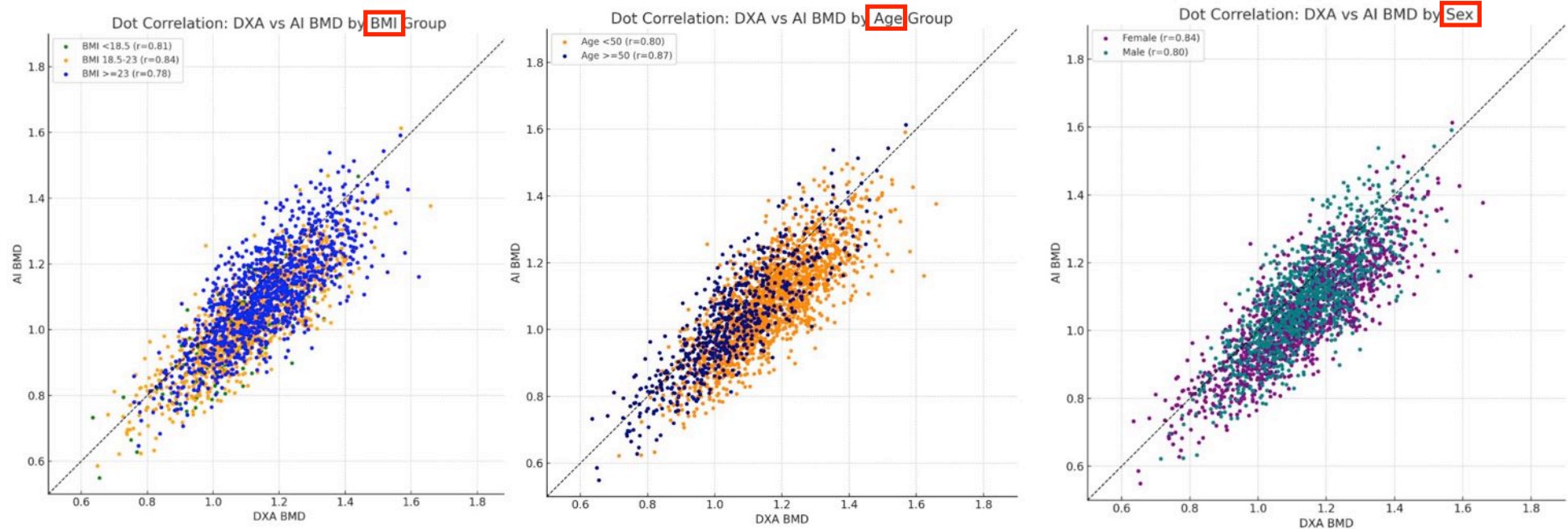


Figure 2A – C. Scatter plots of AI-predicted versus DXA-measured BMD across BMI, age, and sex subgroups.

Scatter plots display the relationship between AI-estimated BMD (y-axis) and reference DXA BMD (x-axis) in the validation cohort. Each dot represents one participant, grouped by (A) body mass index (BMI), (B) age, and (C) sex. The black dashed line indicates the identity line ($y = x$). Pearson correlation coefficients (r) were 0.81, 0.84, and 0.78 for underweight, normal weight, and overweight participants, respectively; 0.80 and 0.87 for <50 and ≥50 years; and 0.84 and 0.80 for females and males. Correlations remained consistently high across all strata, confirming the robustness of AI-derived BMD predictions independent of demographic or anthropometric variation.

Correlation between eBMD and DXA under different BMI, Age and Sex group

Figure 3A–C. ROC curve performance of AI-assisted osteoporosis detection across BMI, age, and sex strata.

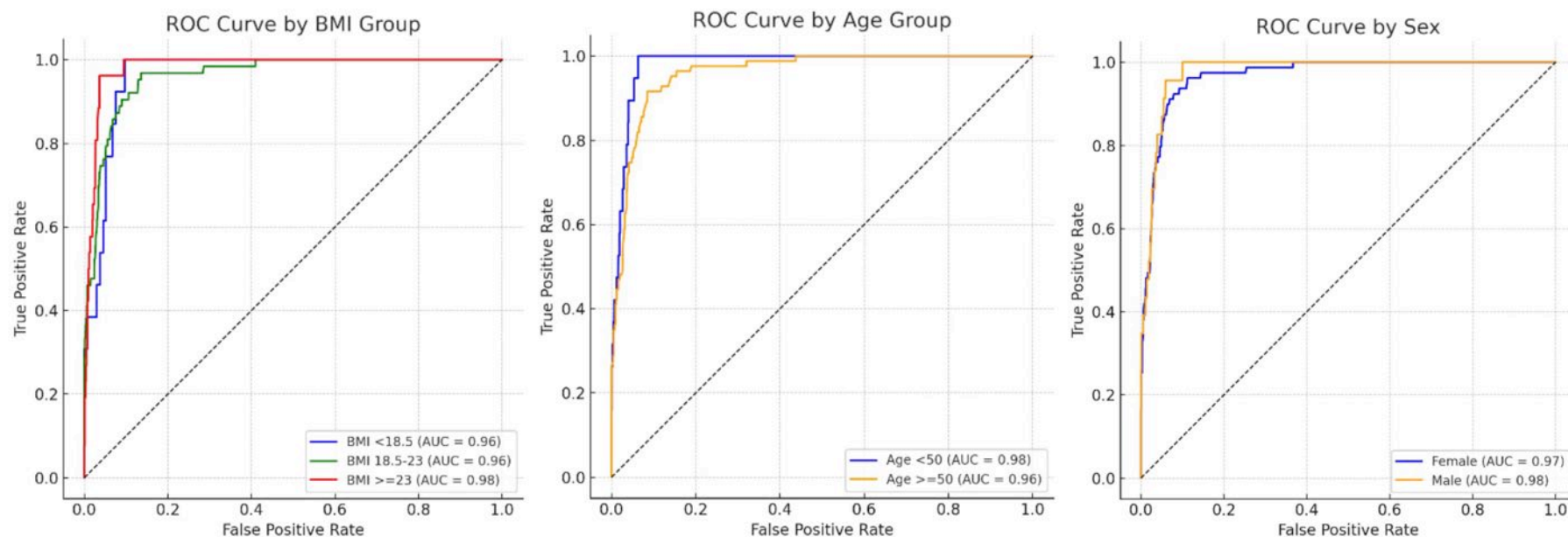


Figure 3A–C. ROC curves of AI-assisted osteoporosis detection across BMI, age, and sex subgroups.

Receiver operating characteristic (ROC) curves illustrate model discrimination between DXA-confirmed osteoporosis/low BMD and normal BMD across subgroups: (A) body mass index (BMI), (B) age, and (C) sex. Dashed lines indicate chance level ($AUC = 0.5$), and shaded areas represent 95% confidence intervals. AUCs were 0.96, 0.96, and 0.98 for underweight, normal weight, and overweight participants, respectively; 0.98 and 0.96 for <50 and ≥50 years; and 0.98 and 0.97 for males and females. High AUCs were observed consistently across all strata, confirming strong discriminatory power independent of demographic or anthropometric variation.

AUC under different BMI, Age and Sex group

Figure 6. Decision curve analysis (DCA) of AI-assisted screening.

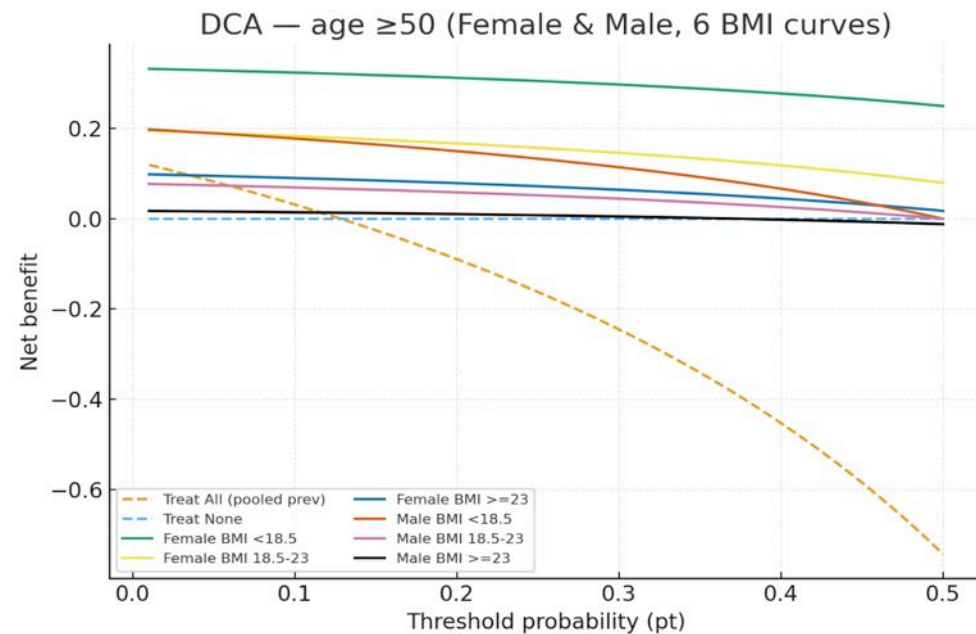
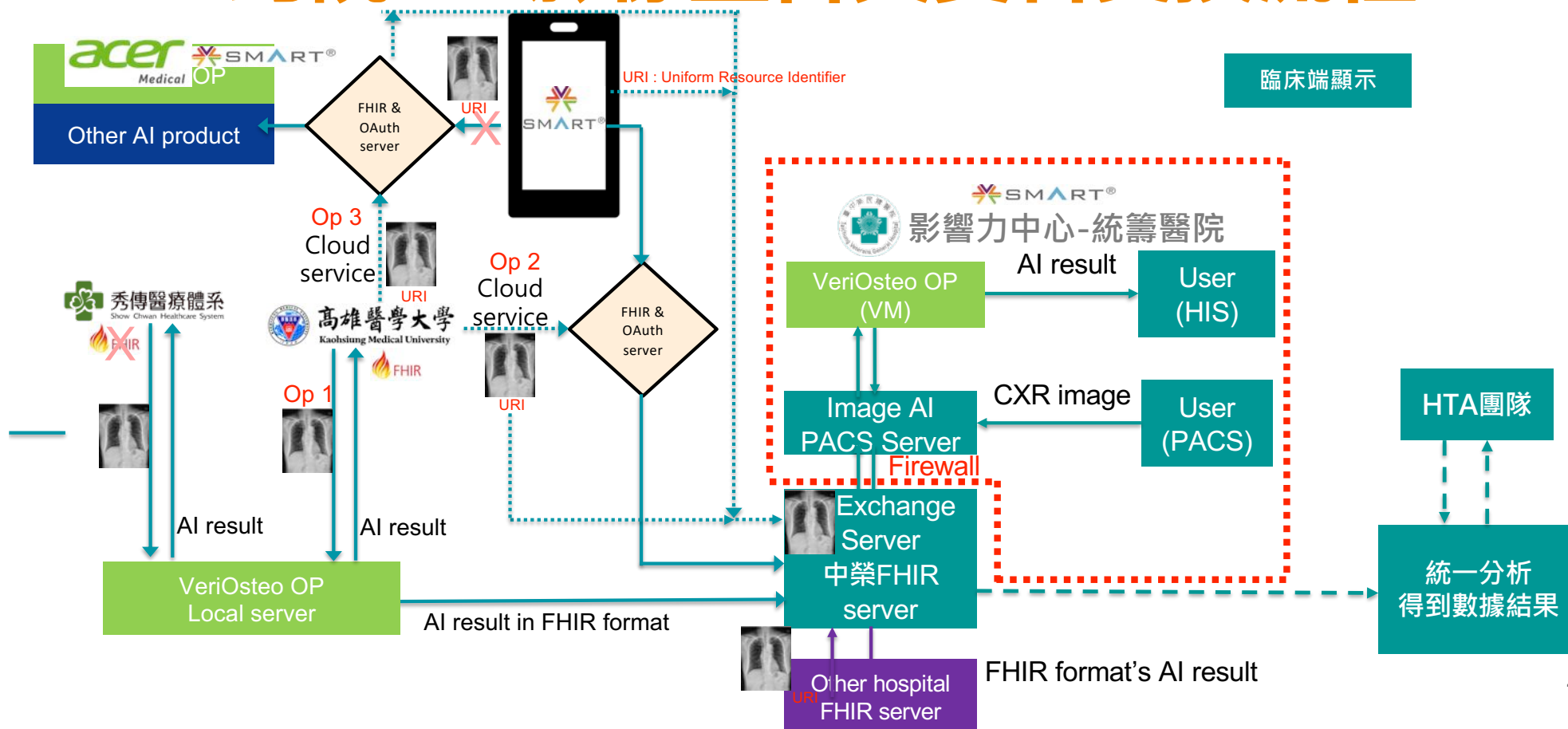


Figure 6. Decision curve analysis of the AI model for osteoporosis detection compared with “screen all” and “screen none” strategies. The y-axis shows net clinical benefit, and the x-axis represents threshold probability. The AI model provided greater net benefit across a wide range of thresholds, supporting its potential utility as an opportunistic screening tool.

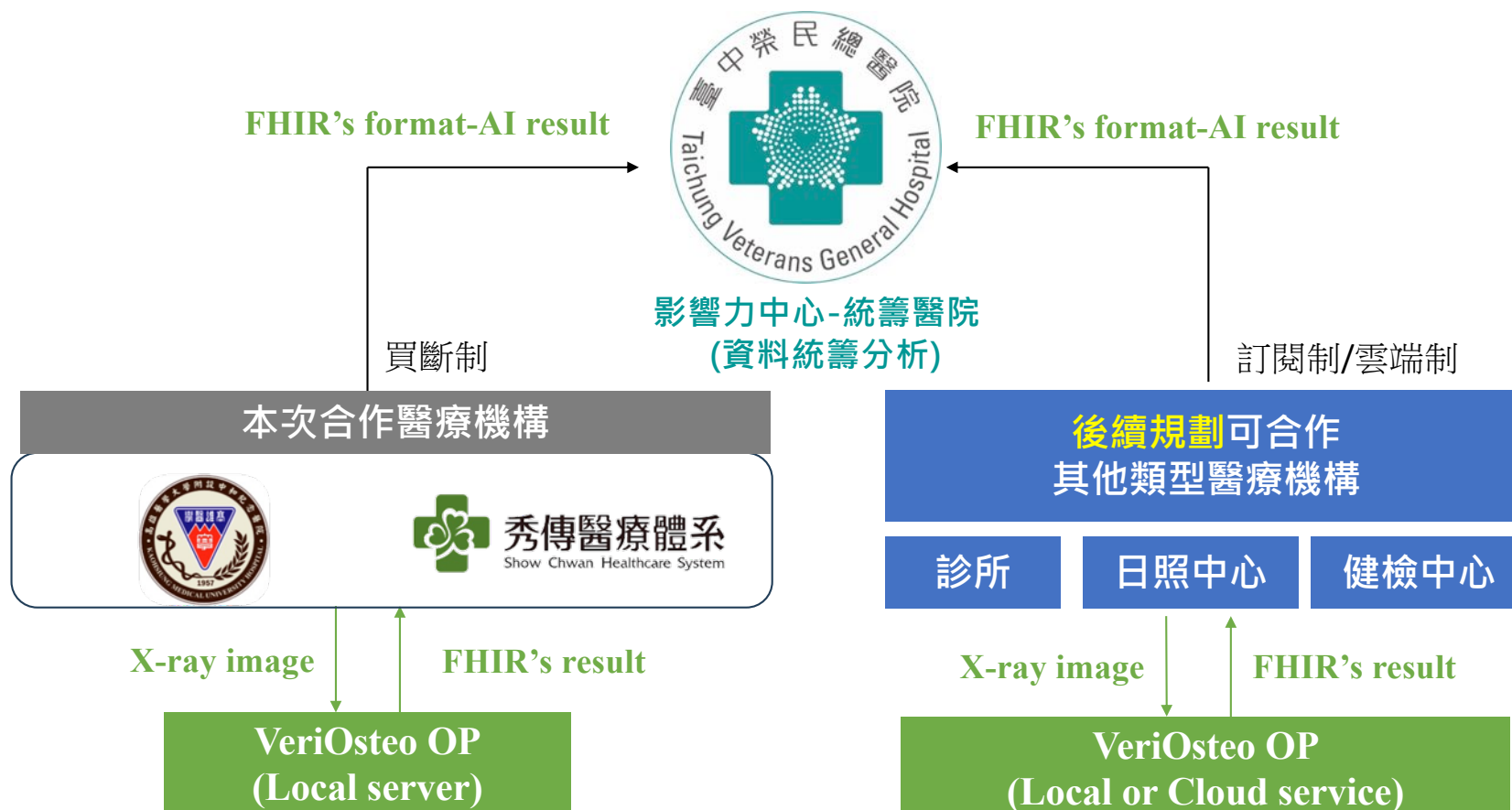
Decision curve analysis of the AI model for osteoporosis detection compared with “screen all” and “screen none” strategies



跨院 AI 影像整合與資料交換流程



SMART on FHIR 部署規劃情境





骨質密度異常篩檢								
AI辨識的骨質密度(BMD)	骨密度 T 值(T)	疑似異常	881	-2.7	疑似異常	0.83011	-2.9	疑似異常
0.83539	-2.8							

Any X-ray will provide answer, even wrong X-ray



X-ray AI classifier

- Classify 111 kinds of X-ray
- 25 major classification
 - Accuracy : 99.95%
 - F1-score : 0.9995
- 111 minor classification
 - Accuracy : 98.65%
 - F1-score : 0.889

X 光影像分類系統

上傳單張影像，使用訓練好的模型進行部位分類。

● 本機推論 · 小分類

1. 上傳 X 光影像



點擊影像可放大查看。

目前模型：☐ 大分類 ☒ 小分類

選擇或拖曳 X 光影像
支援 PNG / JPG / JPEG

已載入上一張影像，可直接切換模型重新推論。

開始推論

2. 推論結果

● 模型推論結果

大分類: chest 小分類: 後前

類別標籤: chest/後前

多模型 Top-1 機率: 93.84 %

多模型 Top-5 類別機率:

類別	機率	數值
chest/後前 Top 1		93.84 %
chest/左_側		0.12 %
chest/前後		0.08 %
spine/L_正		0.07 %
spine/T_正		0.06 %

模型個別 Top-1 機率 (5)

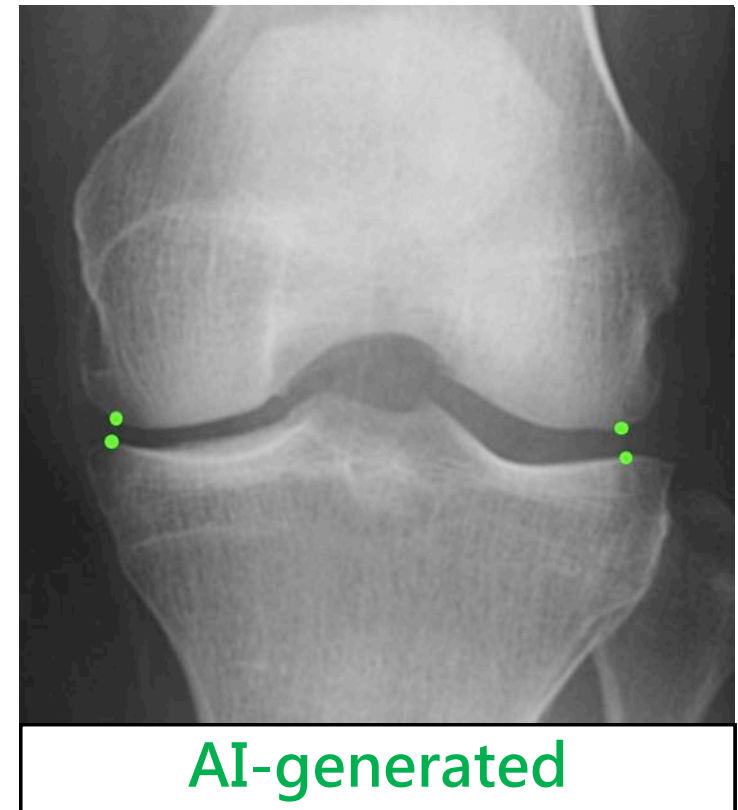
EFFNETV2-S chest/後前 100.0%	DENSENET121 chest/後前 89.6%	RESNET50 chest/後前 89.8%
CONVNEXT-V2-TINY chest/後前 99.5%	INCEPTIONNEXT-TINY chest/後前 90.4%	

* 最終結果由上述模型機率加權平均決定

Auto-labelling of knee joint space

- Blue dot = ground truth
- Green dot = AI-generated

Taiwan Patent
approved



Deep learning model for Kellgren-Lawrence grading of joint space in Knee AP view

Thanks for your attention !

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Orthopedic Department
Smart Healthcare Committee
Clinical Informatics Committee



臺中榮民總醫院
Taichung Veterans General Hospital



中榮官網



中榮FB

Director of Orthopedic Department, Taichung Veteran General Hospital (VGHTC)
Executive Director of Smart Health Committee, VGHTC
Executive Director of Clinical Informatics Committee, VGHTC
Assistant Professor, National Chung-Hsing University (NCHU)
Supervisor of Taiwan Orthopedic Association(TOA) /Taiwan Spine Society(TWSS)

Michael, Kun-Hui Chen